

Organizing for Wind Power: A Guide for West Virginia Communities



Mountaineer Wind Farm, Tucker County, West Virginia

Photo: Evan Hansen

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1. INTRODUCTION

Welcome to the “Organizing for Wind Power” guide! This guide was conceived as a way to support organizing efforts for wind power and was inspired by the Coal River Mountain Wind campaign in Raleigh County, West Virginia. Accompanying resources and information on wind power and Coal River Wind can be found on the “Journey Up Coal River” website at www.journeyupcoalriver.org.ⁱ

The purpose of this guide is to provide information and tools you can use to be an effective community leader for promoting wind power. The guide will provide:

1. **A brief introduction to electricity:** where it comes from, how we consume it, and how much we consume.
2. **Basic information about wind energy:** how it works, what types of wind energy are being developed, how much wind potential exists in West Virginia, and where and how it can benefit your local economy and environment.
3. **Tips, suggestions, and resources for organizing around wind power in your community:** talking points about wind power, tips and ideas for organizing, and suggestions for outreach and advocating for wind power.
4. **A library of resources:** resources to learn more about wind power, what West Virginia is doing to support it, how to contact your local and state representatives to gather their support, and who is already working in their own communities to organize for clean wind energy and new economic development.

While the purpose of this guide is to support the growth of community organizing and support for wind power in West Virginia, the guide contains comparative comments that relate to coal. This guide does not propose wind power as a substitute for coal, but rather as a supplement to the real economic benefits currently provided by the coal industry.

However, as research shows that wind power can generate significant economic and environmental benefits that are equivalent to or greater than those provided by coal in some cases, the guide aims to stimulate more community support for wind power as an opportunity for diversifying the economies of rural coal-producing counties in the state.

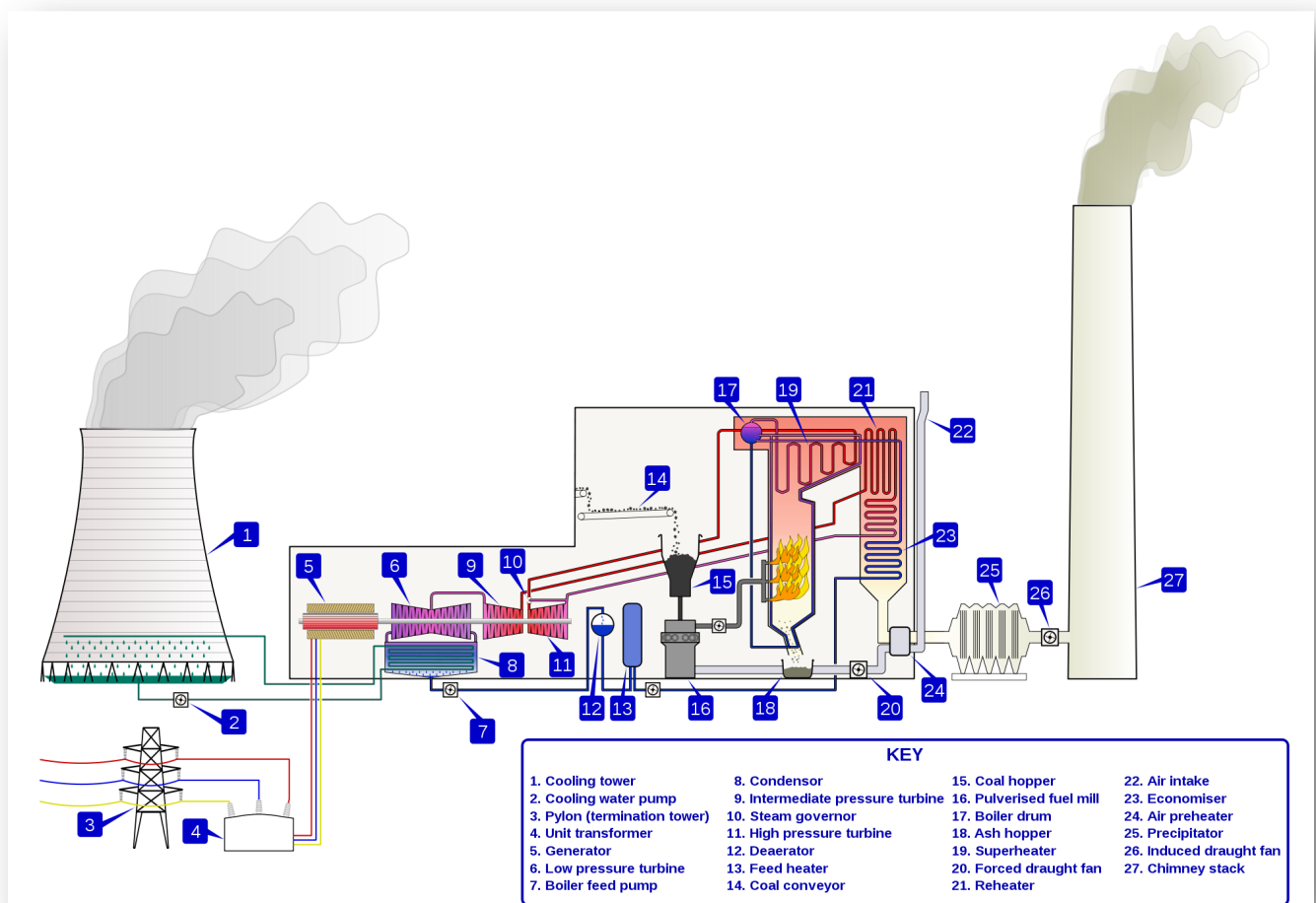
Wind power is growing rapidly as both a source of clean energy and a provider of lasting jobs and tax revenues. It’s also helping to revitalize rural areas across the nation. In West Virginia, three wind farms have been built since 2002,ⁱⁱ and more are under construction and being proposed, suggesting that there are plenty of opportunities for communities to benefit from wind power development. This is because much of the state has excellent wind resources. **Your local and state leaders need to know that your community wants and deserves wind power too!**

By building local support for wind, you are helping to show that there is a growing desire for new, lasting jobs and clean energy in rural communities of West Virginia. You have the power to expand wind power in West Virginia, so get out there and organize!

2. ELECTRICITY AND YOU

Since the late 1800s, electricity has been part of our everyday lives. Today, we use electricity to power our homes, schools, hospitals, businesses, and industry. It's hard to imagine life without electricity, but it's a privilege that many have come to take for granted. Few people understand where their electricity comes from or how it ends up providing the power to light their homes or run their appliances. This is changing, however, as public awareness of issues such as climate change and renewable energy continue to grow.

Most electricity is produced when a fuel is burned and creates heat that turns water into steam; the steam causes a turbine to spin, which generates electricity. In the diagram below, which shows the components of a typical coal-fired power plant, coal is fed into the plant (#14) and burned to boil water (#19 and #21). The steam then turns turbines (#6, #9, and #11) that create electricity (#5). Steam is discharged through the cooling tower (#1), while pollutants are discharged through the smokestack (#27).



A coal-fired power plant

Diagram: Wikipedia, <http://commons.wikimedia.org/wiki/File:PowerStation3.svg>.

The main kinds of fuel used to generate electricity are known as fossil fuels, and include coal, oil, and natural gas. Nuclear power operates in much the same way, using a radioactive element known as uranium. These fuels, since they are finite and will one day run out, are examples of “non-renewable” energy.

“Renewable” sources of electricity include hydropower, geothermal, solar, biomass, and wind.ⁱⁱⁱ These sources are renewable because the fuel never runs out. The water will always flow, the wind will always blow, the earth will release

heat, and the sun will always shine. Each of these technologies also use turbines to generate electricity. With wind power, the wind spins the blades of a turbine sitting on a tower, generating electricity as the turbine spins.

West Virginia can eventually produce a lot of electricity from the wind. This is important because research suggests that the best coal will soon run out and coal production will decline substantially,^{iv} meaning that state and local leaders should begin finding other ways to provide jobs and energy for West Virginia's citizens. Each of the renewable resources listed above are available in our state; we just have to develop them.

Finally, electricity is a privilege that we should not take for granted. Unfortunately, a lot of electricity is wasted, and we can all do more to conserve it. Equally important is that we are mining and burning fuels that produce a lot of pollution that gets into our bodies, streams, and air; the more we use, the more pollution we create. This is another reason why developing clean sources of energy like wind and solar power are important. Although developing renewable energy requires some extraction and processing of resources, once the technology is developed and installed, the generation of electricity is clean because you don't have to add more fuel to keep it going. Also, the resource is free because you don't have to pay for the wind or the sun.

In this way, wind power offers a number of benefits that will have a positive impact for the state. It provides the opportunity to diversify the way we produce energy, reduce pollution to our air and water, and provide jobs that will last as long as the wind blows.

The next chapter will tell you how West Virginia can benefit from developing its wind resources, and it will provide you with the information you need to talk about and advocate for wind power in your community.

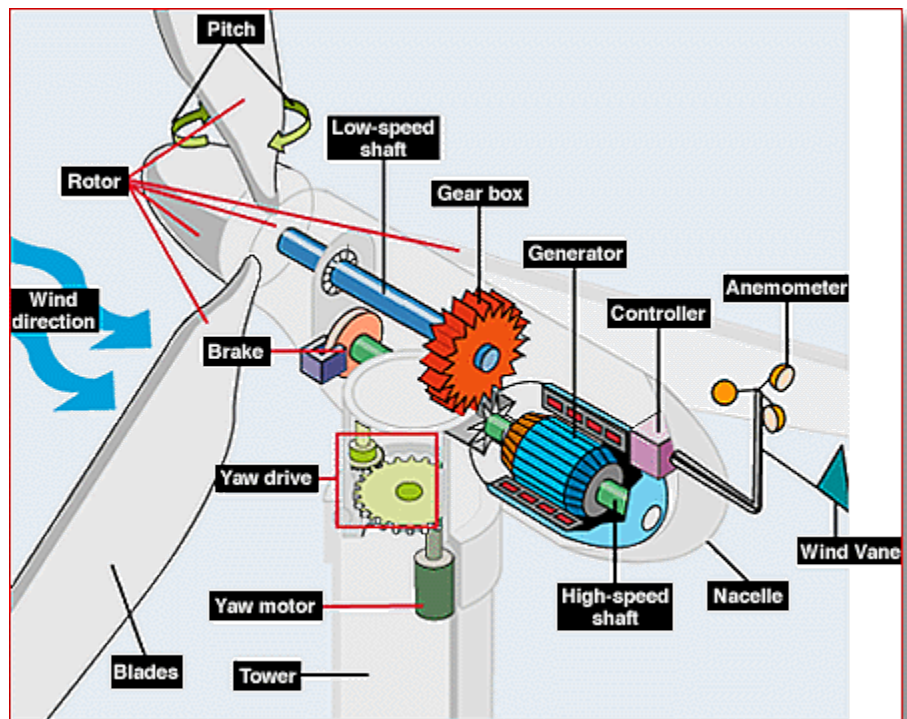
In West Virginia, over 95% of our electricity comes from coal, while only 1% comes from wind power. This makes sense, because West Virginia's biggest resource for energy is coal. In fact, the average home in West Virginia consumes about six tons of coal for electricity each year!

3. WIND POWER FOR WEST VIRGINIA

To organize effectively for wind power and communicate the positive benefits of wind, it's important to understand how wind turbines work, the ways in which wind can be developed, and how wind power can benefit West Virginia.

3.1 How wind power works

Wind turbines convert the energy contained in the blowing wind into electricity. More specifically, wind passing through the blades of a wind turbine spins a shaft, which connects to and spins a generator that makes electricity. In order to capture as much wind energy as possible, wind turbines are built up to 450 feet high. The higher the turbine, the more energy it will produce because winds blow faster at higher altitudes.



Wind turbine components

Diagram: Wikipedia, http://en.wikipedia.org/wiki/File:EERE_illustration_large_turbine.gif.

Wind turbines can be used to produce electricity for a single home or building, or they can be connected to an electricity grid for more widespread electricity distribution. Transmission lines allow for the electricity from a wind farm to be distributed to a greater number of homes, businesses, and schools over a broad area.

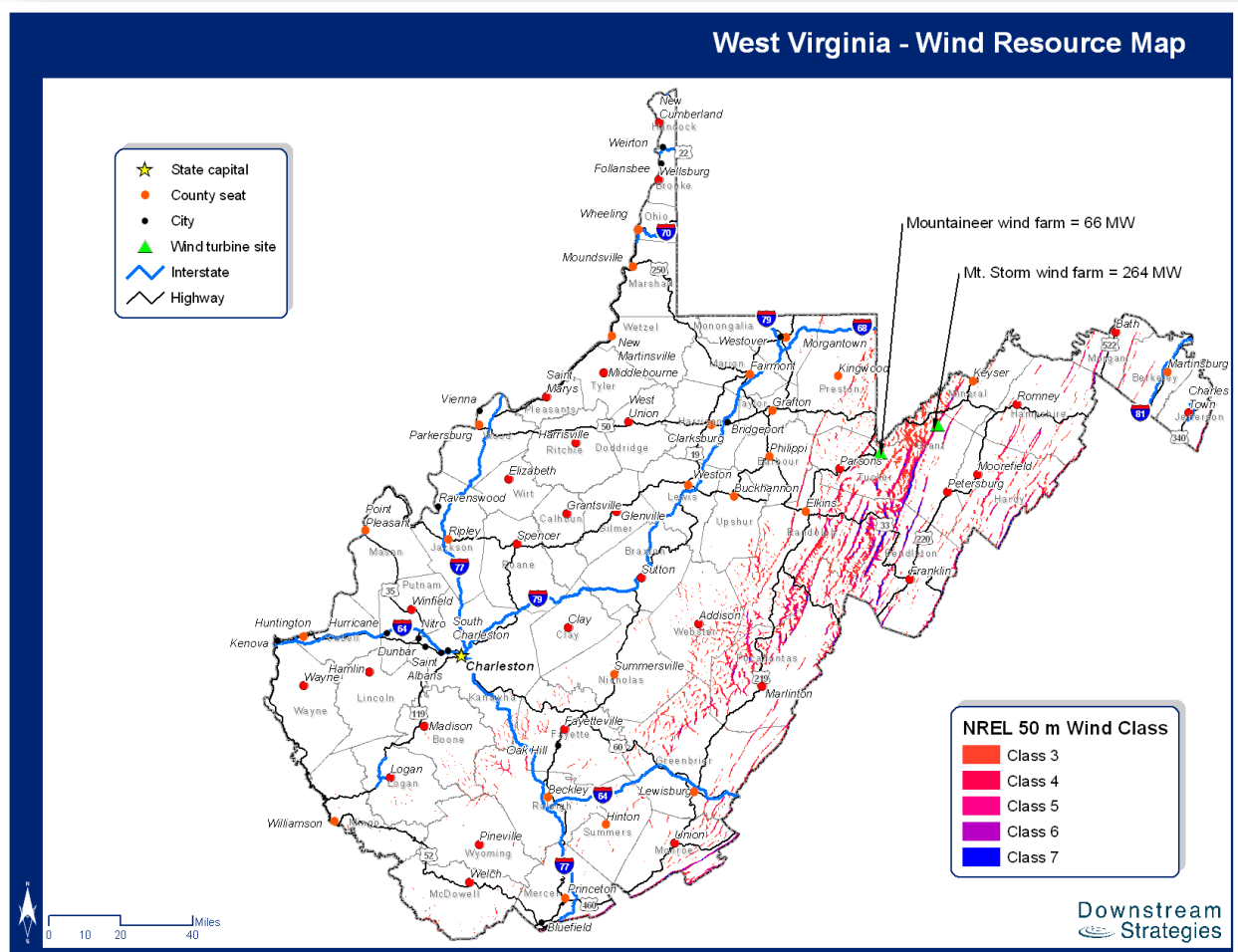
3.2 Wind energy potential in West Virginia

A lot of wind blows across West Virginia's mountain ridges, but little of it has been harnessed for wind power. About the time that the first wind farm was constructed in West Virginia, a study was conducted for the state Division of Energy. The 2001 study concluded that there was enough wind on private lands (outside of state and national parks) to provide about 30% of all the electricity consumed in West Virginia using industrial turbines: 3,830 megawatts (MW).^v The study concluded that even more wind power—another 8,500 MW—could be generated by smaller-scale wind development.^{vi} That's a lot of electricity.^{vii} It's likely that as wind turbines get bigger and more efficient, and as storage technologies improve, even more wind energy can be produced.

The following map divides wind speeds into classes. The higher the class, the more wind is available to generate electricity. Areas designated as Class 4 and above are great for wind development, and even mid-Class 3 areas can be developed. There is good wind potential in many parts of West Virginia, including southern counties like Raleigh and Boone County.

Baseload Generation

Wind power is not yet used for “baseload” generation, meaning it can't provide electricity to meet the average everyday demand for electricity. This is because the wind is not always blowing at a particular site (although it's always blowing somewhere), and so wind turbines don't always produce electricity. With advances in storage technologies such as batteries, perhaps one day that will change. Wind-generated electricity is used today mostly as a supplemental source of electricity, however, and when the wind is blowing, the electricity can be used.



Wind power is already being rapidly developed in West Virginia. So far, three West Virginia wind farms generate a total of 330 MW.^{viii} Another 461 MW are expected to be developed by the end of 2010 in Grant, Greenbrier, Randolph, and Barbour counties, for a total of 791 MW.^{ix} Other wind farms have been discussed by developers in Grant, McDowell, Mineral, Nicholas, Pendleton, and Tucker counties.^x Residents of the Coal River Valley in Raleigh County have also proposed a 328 MW wind farm as an alternative to a mountaintop removal coal mining operations.^{xi}

Wind power is growing in West Virginia, but only in a few areas, and many residents of the coalfields are beginning to want wind power in their communities too!

3.3 Types of wind development

Commercial-scale wind turbines sometimes receive quite a bit of local opposition due to the fact that they are large, industrial operations. For instance, the largest wind turbines being put on West Virginia's ridges are up to 450 feet tall! That's one-and-a-half football fields high. Big wind farms aren't necessarily bad, though; properly sited they can produce a vast amount of clean energy with little environmental impact compared with coal mines. However, this is not the only way wind can be developed.



Commercial-scale wind turbine in Tucker County, West Virginia

Photo: B. Mark Schmerling, Schmerling Documentary Photography.

3.3.1 *Small wind*

For powering a home, school, or business, small wind turbines are the most common choice. They can range from less than 1 kilowatt (kW)^{xii} to around 100 kW or more, so they generate less electricity than a big 2 MW wind turbine. But they are also being used to provide energy to fewer people or locations, so less electricity is normally required. They are also far less expensive than big turbines, and so are more affordable for your home, and they can be used to supplement your electricity consumption, helping to reduce the amount you pay each month to the power company for electricity. In months that you generate more electricity than you use, you can get a credit toward free electricity from the power company the next month. This is known as “net metering.”

Finally, small wind turbines don’t normally receive opposition, because no more than a few are developed on any single location, and they’re owned by and provide electricity to local residents or businesses.

Net Metering

Net metering serves as an important incentive for community and individual investment in renewable energy generation. Net metering enables customers to use their own wind turbine or solar panel to offset their electricity consumption each month by allowing their meters to turn backwards when they generate electricity in excess of what they use. In many states, homes or businesses are paid by their utility for the excess electricity they generate. The current net metering law in West Virginia doesn’t require the power company to pay you for your extra electricity. Instead, you can receive a credit against next month’s power bill.



Construction of a small wind turbine

This 10 kilowatt turbine could power an average home.

Photo: Power In My Back Yard (PIMBY), www.getpimby.com.

3.3.2 *Community wind*

Community wind is defined as wind development that leverages local distribution grids to the environmental and economic benefit of the local community, while providing investment opportunities for community members to become financial partners or owners.^{xiii} Community wind projects are locally owned by farmers, investors, businesses, schools, utilities, or other public or private entities and they optimize local benefits.

The key feature is that community members have a significant, direct financial stake in the project beyond land lease payments and tax revenue. Projects may be used for on-site power or to generate wholesale power for sale. They are different than small wind installations, in that they are typically composed of large wind turbines 1-2 MW in size, and the wind farms typically range from 5 to 30 MW. One great example of a proposed community wind project is the Angel Wind Project in Monroe County.^{xiv}

Some projects use smaller wind turbines of 100 kW to 1 MW. But larger turbines mean more revenue because the electricity that isn't consumed locally is sold back to the electric utility by the owners. This means that communities that install larger turbines can get their money back faster than with small wind turbines. Community-owned wind projects may provide the best and more beneficial option for wind development, because the turbines, the electricity, and the revenues are owned—at least in part—by the community!



Community-owned wind turbine in Hull, Massachusetts

Hull Wind II is a Vestas V80 1.8 MW community-owned turbine that went online in 2006. The turbine provides Hull, a town of 10,500, with 9% of its electricity. The electricity is run through the town-owned electric utility, which makes this a community wind project.

Photo: Andrew Stern, www.hullwind.org.

Community wind contributes more to the local economy than privately owned wind farms by keeping investment dollars and revenues in the local community, and they tend to generate more jobs as well. A 2009 study compared the average employment impacts of community wind projects with the impacts from the first 1,000 MW of corporate-owned wind developed in Colorado and Iowa, and found that construction-period job impacts are as much as 3.1 times higher for community wind than for corporate wind, and operations-period job impacts are as much as 1.8 times higher. The study concluded that:

“Ultimately, wind projects are a source of jobs and economic development, and community wind projects are shown to have increased impact both during the construction and operations-period of a wind power plant. The extent of increased impact is primarily a function of local ownership and return on investment. As such, policies that prioritize higher levels of local ownership are likely to result in increased economic development impacts.”^{xv}

Websites

American Wind Energy Association (AWEA)
www.awea.org

National Renewable Energy Laboratory (NREL)
www.nrel.gov/wind

Windustry
www.windustry.org

US Department of Energy, Wind Powering America
www.windpoweringamerica.gov

Renewable Energy Policy Project (REPP)
www.repp.org/wind

Appalachian State University, Renewable Energy Initiative
www.rei.appstate.edu

Energy Information Administration (EIA)
www.eia.gov
www.nrel.gov/wind/pubs_issues.html#impacts

Community-owned wind farms are growing all over the nation, and West Virginia—with its vast wind resource and need for new economic opportunity and local ownership—is the perfect place for community wind projects to be developed. See the “Documents and other information” box for websites where you can learn more about community wind and how to get it for your community.

3.4 The economic benefits of wind power for West Virginia

3.4.1 *Taxes for the state*

Wind power has the potential to provide significant economic benefits on both the state and local level. On the state level in West Virginia, wind farms generate business and occupation taxes. For instance, for every 100 MW of wind development, the state receives nearly \$275,000 per year, which can then be put towards education, improvements for roads, and hospitals.^{xvi} The state also receives corporate income and personal income taxes. The more wind that is developed and the more jobs that are created, the greater the tax benefit to the state will be.

3.4.2 *Local economic benefits: Property taxes and landowner revenues*

On the local level, wind power generates significant tax revenues in the form of property taxes, landowner revenues in the form of lease payments, and jobs in the construction, operation, and maintenance of wind farms.

Using the model of the wind farm proposed for Coal River Mountain in Raleigh County, it was found that the 328 MW wind farm would generate an average of \$1.74 million per year in county tax revenues, forever,^{xvii} which is comparable to what the county receives in a record coal severance tax year such as in 2008.^{xviii}

Landowners who lease their land for wind development might earn over \$10,000 per turbine, or about \$5,000 per MW per year.^{xix} Estimates for West Virginia provided by the National Renewable Energy Laboratory (NREL) were about half that: \$2,700 per MW or \$5,400 for each 2 MW wind turbine.^{xx} The true value would depend on the amount of electricity generated and sold and on the specific agreement between the landowner and the wind developer.

3.4.3 *Local economic benefits: Jobs*

As for jobs, the model Coal River Mountain wind farm could be expected to create 362 direct and indirect construction jobs during the two-year construction of the wind farm, and another 48 permanent direct and indirect jobs related to the operation and maintenance of the wind farm.^{xxi} According to NREL, between 100-200 construction jobs and 6-10 operations and maintenance jobs could be created for every 100 MW of wind development.^{xxii} Using NREL's estimates, developing all 3,830 MW of West Virginia's large-scale wind potential on private lands would create approximately:

- 6,600 direct and 5,600 indirect and induced construction-related jobs, for a total of about 12,000 new jobs, and
- 1,000 direct and 600 indirect and induced operations and maintenance jobs, for a total of 1,600 new, permanent jobs.

3.4.4 *Diversifying the local economy*

Of greatest importance, however, is that the development of wind farms and possibly a wind industry would diversify the local economy. The development of a local wind industry with manufacturing capabilities would increase the local employment benefits of wind power dramatically. In fact, NREL estimates that between 1,000 and 5,000 new manufacturing jobs could be created in the wind industry in West Virginia by 2030.^{xxiii} By comparison, the study conducted for Coal River Mountain estimated that a local wind industry that manufactures the towers, turbines, and blades for Coal River Mountain alone would provide 1,750 jobs directly and a total of 3,831 jobs per construction period, including the indirect and induced employment effects.^{xxiv} These jobs could then be preserved over the long term with additional wind development.

Wind development can coexist with underground coal mining, with sustainable timbering, and with the development of other local economic opportunities. Wind power also provides steady jobs that aren't sensitive to boom-and-bust cycles. The jobs exist as long as the wind blows.

Finally, the addition of new sources of revenue at the county level provides extra income that can be reinvested in the community—for instance, to improve schools, to build new local health care centers, to build playgrounds and parks, and to pay for cleaning up streams, developing other economic opportunities such as tourism, and providing other needed services such as fire and rescue.

3.5 Environmental costs and benefits of wind

Wind power comes with both environmental costs and benefits. Wind farms in West Virginia may be developed on ecologically sensitive ridgelines, threatening birds and bats. Construction also requires some land and forest clearing for access roads, which reduces wildlife habitat and can cause forest fragmentation. Finally, road construction may result in sediment runoff, which, if improperly controlled, may harm streams. Each of these impacts may be minimized, however, and are relatively small when compared to the impacts of coal mining and coal-fired power plants.

Wind power comes with environmental benefits too. Compared to surface coal mining, relatively little timber must be cleared for roads and turbines, and little to no blasting is required. Wind farms do not consume water resources, and except for sediment runoff from access roads, they don't generate water pollutants. They don't release air pollutants such as sulfur dioxide, nitrogen oxides, carbon dioxide, or mercury. Communities may also continue to use the lands and forests surrounding the wind turbines in order to maintain traditional uses of the land. Overall, the development of wind power generates significant environmental and health benefits when compared to conventional energy sources. Now that you know all of this....

Other wind organizing campaigns and organizations in West Virginia

Coal River Mountain Watch, Coal River Wind Project

www.coalriverwind.org

www.crmw.net

(304) 854-2182

Lorelei Scarbro

lolelei@coalriverwind.org

Julia Sendor

julia.b.sendor@gmail.com

Ohio Valley Environmental Coalition, JOBS Project

www.jobs-project.org

www.ohvec.org

(304) 784-0864

Eric Mathis

thejobsproject@gmail.com

Jenny Hudson

jennysuehudson@gmail.com

Aurora Lights, Journey Up Coal River Project

www.JourneyUpCoalRiver.org

(304) 698-5160

Jen Osha

jenosha@auroralights.org

Student Environmental Action Coalition (SEAC),
West Virginia Youth Action League (WV-YAL) Project

www.seac.org/wvyal

(304) 886-3389

Danny Chiotos

trouserdude@gmail.com

4. GET OUT AND ORGANIZE

Wind power is a fairly new industry, and because of this, a lot of people and political leaders are uncertain how they feel about wind. This is especially true in areas such as the coal counties of West Virginia, where there isn't any wind development yet, and where the coal industry currently serves as the economic engine. New industries are possible though, not necessarily as a replacement for coal, but in addition to coal. Once people understand that, they will be more likely to accept and support wind power. But it's up to you to help make that change, and the best place to start is in your own community and county.

This chapter provides talking points about wind power, tips and ideas for organizing, suggestions for outreach and advocating for wind power, and information and resources for purchasing or developing wind power for your home, community, business, or school. Remember that this guide provides merely an overview, and that there are vast resources available to you if you would like to learn more.

You know your community better than anyone. You spend time with your neighbors, you know their kids, and you help each other out. Because of this, only you will know how to talk to them about something as new as wind power. However, the following section offers some suggestions.

Documents and other information

ENERGY

Wind power tutorial

<http://awea.org/faq/index.html>

Wind energy basics

www.nrel.gov/learning/re_wind.html

Windustry, Community Wind Toolbox

www.windustry.org/CommunityWindToolbox

ECONOMIC IMPACT

Economic impacts of community wind

www.nrel.gov/docs/fy09osti/45555.pdf

Wind for rural economic development

www.windustry.org/wind-energy-for-rural-economic-development

www.nrel.gov/wind/pdfs/39099.pdf

Wind energy and economics

www.repp.org/wind

EDUCATION

Wind Energy Teacher's Guide

www.awea.org/pubs/documents/TeachersGuide.pdf

Wind Energy for Schools

www.windpoweringamerica.gov/schools.asp

KidWind Project

www.kidwind.org

POLICY

Wind Energy Guide for County Commissioners

www.windpoweringamerica.gov/filter_detail.asp?itemid=1386

Database of State Incentives for Renewable Energy (DSIRE)

www.dsireusa.org/index.cfm?EE=0&RE=1

AWEA "New Wind Agenda"

www.newwindagenda.org

Small Wind Turbines: A Menu for State and Local Governments

http://awea.org/smallwind/pdf/Policies_to_Promote_Small_Wind_Turbines.pdf

Community Wind: A Review of Select State and Federal Policy Incentives

www.flaginc.org/topics/pubs/arts/CommWindAug06.pdf

OUTREACH

Model letters to neighbors, elected officials, and newspaper editors

http://awea.org/smallwind/toolbox2/sample_letters.html#intro

Small wind development guide

http://awea.org/smallwind/toolbox2/drawer_1_installation.html

4.1 Suggested talking points when organizing for wind power

One thing to remember is that you learn more from others, you strengthen your effectiveness by listening, and you empower others by enabling them to talk about wind power themselves. This is the best way to organize. As is often said, the goal of organizing is to work yourself out of a job by turning others into community leaders. That said, here are some suggested talking points about wind power:

1. **Wind energy provides jobs for your community.** Depending on the amount of wind power developed, a wind farm can create tens or hundreds of jobs that will last as long as the wind blows. These are safe, well-paid jobs, and they can be union jobs, or even jobs within a cooperative if you're the ones who own the wind turbines. They are also jobs that you and your neighbors are used to, including everything from management positions, sales representatives, and contractors to equipment manufacturers, truck drivers, road builders, and heavy machine operators. These jobs will still exist even if coal production declines.
2. **Wind power generates local tax revenue.** Wind farms can help revitalize rural areas by providing new sources of tax revenue that can be spent on schools, roads, and health care centers. The tax revenues will not waver from year-to-year because wind power is not susceptible to boom-and-bust cycles like coal mining. As long as a wind farm exists, it will provide tax revenues.
3. **Wind farms generate revenues for landowners.** In much the same way as leasing coal to a coal company, leasing land to a wind developer can earn a landowner thousands of dollars per turbine, forever. The amount of revenues will vary according to the size of the wind farm and the lease agreement. The benefits can be even greater for community-owned wind projects.
4. **Wind energy stabilizes electricity prices, saving money over time.** Once a wind farm is built, it doesn't require any fuel. It only requires the wind, which is free. Costs for conventional energy sources such as coal, however, are cyclical and may rise over the long term, so wind power offers price stability and safeguards you from paying higher prices for electricity in the future. If the electricity is run directly to your home, you don't have to pay a utility company for your electricity.
5. **Wind energy is clean energy.** Wind farms don't produce any pollutants that can harm your lungs or contaminate your drinking water. They don't contribute to global climate change or add smog to the air. They also have relatively little impact on the land, and they don't prevent you from using the land as you always have, such as for hunting or fishing.
6. **Wind energy is renewable energy.** The wind will always blow. It will never run out.
7. **Wind is plentiful in West Virginia.** There are a lot of high mountain ridges in the state, meaning there is a lot of wind blowing through the mountains. A lot of wind power can be developed, and a lot of jobs created.
8. **Wind needs help in West Virginia.** Just because the wind blows doesn't mean it will be turned into energy. There aren't many policies or incentives on the local or state level that support the development of wind power. Because of this, you should ask your elected officials to pass legislation that requires wind power to be part of the state's electricity generation mix, and that provides financial incentives for all types of wind development: commercial-scale, small-scale, and community wind. This will increase the chances that you can get wind power for your community and state.
9. **Everyone deserves new jobs and clean energy.** Wind power, as well as other renewables, are being developed all over the United States, creating new jobs and manufacturing opportunities for rural communities. Don't we all deserve those jobs? Don't we all deserve clean energy and economic stability?
10. **Why Not Wind?** With all of these benefits, you can ask your friends, neighbors, and elected officials: Why not wind? There is little reason to oppose wind power in relation to the potential benefits of developing it. Prepare yourself with the knowledge necessary to talk about wind power, so that when someone has concerns even after you've shared what you know, you can say: Well, why not?

You're now equipped with some useful talking points about wind power. These are by no means the only ways to talk about wind. Find out what your neighbors and co-workers are concerned about, and what questions they have, and talk to them about it. Then get them involved too! There are many ways to build support for wind power, and the following section offers some ideas for doing that.

4.2 Tips for organizing

Organizing isn't always easy, especially when you're introducing a new idea. The first thing to remember is not to be afraid of talking about wind power. The very fact that you support it might be enough to convince others to listen and ask questions. Also keep in mind that if someone says no or opposes you, don't argue. Accept their right to their opinion and move on. Other than that, here are some reminders and ideas for being an effective organizer for wind power:

- **Listen.** The first rule of organizing is to listen more than you speak. You can't address people's concerns or answer their questions if you're doing all the talking. Introduce the idea of wind, share what you know, and let others do the rest of the talking. The better you understand the person you're talking to, the better you will be able to communicate and work with them.
- **Get to know your community.** Spend time with your neighbors and friends. You don't always have to talk about wind power, either.

Feeling comfortable around others, especially someone you've just met, is important for being able to connect with them and have an open conversation. Don't be afraid to go door-to-door (although only do so if you're comfortable going door-to-door in your community). Also, once you've developed a relationship with one individual or family, ask if they will introduce you to their friends or neighbors. Having someone you already know introduce you can help you to get to know new people.

- **Encourage leadership and involvement.** Your job is to identify and develop community leaders, and to support them as much as you can. Never do something for someone that they can do for themselves, and be sure to involve others in every aspect of the decision-making from the very beginning. Let others take the lead if they show that they're willing. The whole point is to grow support, and everyone works better if they have a sense of ownership. Once a group of community members has gotten involved, it might be good to begin splitting up responsibilities. This ensures that everyone stays involved, and it makes campaigning more effective.
- **Don't get discouraged.** Nothing happens quickly, especially when you're introducing something new. Your political representatives were likely elected by folks who depend on and believe in the things they already know, so it's going to be hard to get your politicians to advocate for something new like wind power. This is why organizing your community, and expanding your efforts into other communities, is so important. The more voices there are asking for wind power, the more chance you will have to get your elected officials to support you. Just don't give up, and be sure to recognize and celebrate victories when they happen.
- **Finally, be accountable and stick to your word.** Make sure that you are on time to the meetings that you organize, and if you have to cancel, give participants plenty of notice. Try not to promise anything that you are not sure can be delivered. Sometimes it helps to keep the steps small so that everyone can see that something has been accomplished, even if it is just a mailing list or a letter to the editor.

To find out what's happening with wind in West Virginia

West Virginia Wind Working Group

Patrick Mann

(304) 599-2641

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www.energywv.org/community/WindConferences.html

West Virginia Division of Energy (and Public Energy Authority)

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www.energywv.org/community/eep.html

West Virginia Environmental Council

Don Garvin

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www.wvecouncil.org

Marshall University, Center for Business and Economic Research

Christine Risch

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4.3 Things you can do: Organizing efforts

Using the principles and talking points outlined in the previous sections, you're now ready to engage your community. This section will give you some ideas for generating support and advocating for wind energy. This is not a complete list. Only you will know the best way to connect with your community and elected officials, as well as what your resources are in terms of time, energy, and money.

- **Plan community events and get-togethers.** One way to bring the community together to get to know each other and talk about wind is to organize a community event. This could be a meeting, a cook-out, a field trip, or a shared project such as a stream clean-up or walk into the woods. Such events are more effective when they're creative and fun for everyone.
- **Pass out flyers and information.** A quick way to spread information or get people interested is to create informational flyers that spark people's interest. These can be passed directly around your community, posted in a church or post office, or even printed in a local newspaper. They are good for sharing information or inviting people to events.
- **Develop a team of organizers.** Once you begin gathering support in your community, make it official. Give the group a name, or begin scheduling regular planning meetings in order to keep everyone involved. Once you are joined by one or more people, begin splitting up tasks among the group.
- **Develop a list of principles and talking points.** Getting everyone's input, begin putting a plan together. Discuss and agree upon a set of principles and a mission statement that will serve as the basis for your efforts. What are you supporting and why? What are you opposed to? Out of these principles should come a set of talking points that will be used continuously throughout your campaign.
- **Develop your goals and strategy.** Come up with specific goals that align with your mission statement. For example, if your mission is to get a community-owned wind farm developed, then your goals will be defined by that mission. However, if your mission is to support wind power more broadly across the state, then your goals might include lobbying state legislators for a specific piece of legislation. Once you've defined your goals, then you should begin developing a strategy for achieving them.
- **Generate supporting materials.** Handouts are always useful. Begin putting together a document with your mission statement. Generate flyers that can be used continuously. Create a petition and begin gathering signatures supporting your campaign. Put together a list of community members and their contact information, and then do the same for your state and local leaders. Make buttons, t-shirts, yard signs, and other items that show support for wind power. These are just a few ideas, but the better prepared you are, the better you can prepare others, and the more effective you all will be.

State and local officials and agencies

West Virginia Economic Development Authority
www.wveda.org
(304) 558-3650

West Virginia Development Office
www.wvdo.org
(304) 558-2234

West Virginia Division of Energy
www.energywv.org
(304) 558-2234

Appalachian Regional Commission
www.arc.gov/index.do?nodeId=13
Ralph Goolsby, State Program Manager
rgoolsby@wvdo.org
(304) 558-2234

County Commissioner's Association of WV
www.polsci.wvu.edu/wv
(304) 345-4639

4.4 Things you can do: Outreach and advocacy

Once you're ready to take the next step, it's time to get out and advocate for wind power! One thing to remember is that you should always be aware of opportunities to take advantage of. For instance, perhaps your County Commission is holding a public event, or a state agency is holding a meeting related to energy. These are good opportunities for your community to learn what is happening and to ask questions or make your voices heard. Other than that, here are just a few suggestions for outreach and advocacy:

- **Write letters.** There is no easier way to make your voice heard than to write letters to specific targets. Letters to the editor of your local and state newspaper are an important tool in advocating for wind. Only a couple of years ago, few people were talking about wind power, but now it is becoming a common topic in conversations about energy in the state. The public and your representatives need to know that more citizens are thinking about wind and wanting clean energy for their communities.
- **Contact your representatives.** You can also write letters to your elected officials and request a meeting with them in person. After all, they depend on your vote. Tell them what you think, and let them know why you want wind power and what they should be doing to support you. When you have a meeting, bring as many community members as you can. Also, introduce specific policies to your representatives that have been implemented in other states, and let them know that you're willing to work with them to introduce similar policies. Target potentially "friendly" representatives who show that they are willing to help.
- **Invite speakers and elected officials to visit your community.** You may not receive a positive response, but inviting key officials or people who have experience with wind power may be an effective way to quickly earn support for wind. Speakers who know about wind power will help by sharing their experiences with your community. Having your elected officials visit will help remind them of who they are representing and will serve to personalize your efforts.
- **Meet with appropriate agencies.** Ask for a meeting with your state and local economic development authorities and energy agencies. Talk with them about wind power and why you support it, and let them know how many other residents you represent. Bring as many community members as you can to the meetings, and ask them for their help in providing new wind jobs for your area.
- **Plan supporting events.** Make your voice heard. Organize a public forum to discuss wind power with other local residents, or put together a press conference to let the public and your local and state government know what you support. Invite the media; they are always interested in new efforts and ideas. Other ideas include taking community field trips to see wind power in action, and attending conferences or other wind-related events.
- **Work with local schools.** Education is a vital tool for growing support for wind power and other renewable energy opportunities. Work with local schools of every grade level to develop and incorporate curricula and activities focused on renewable energy. Offer to come and speak in the classrooms, or set up after-school activities for children to participate in.
- **Develop a network of support.** Contact other communities and groups who are working for the same goals that you are, and begin developing a united effort. There are a lot of groups now talking about and working for wind, many of which are listed in this guide. Let them know they have your support, and ask for theirs.
- **Again, never give up!** If you hit a wall, find a way around it. If your elected officials don't listen, vote for someone who will! Just don't give up; make your voice heard. And remember to support those who support you, especially the people who make up the community you live in.

Hopefully, this guide will prove useful to you in your efforts to educate yourself and others about wind power for your community and for West Virginia, and to grow support for wind and for a diversified economy in the coalfields. Your community deserves clean energy, and it deserves new economic opportunities, but it is up to you to show your elected officials that the citizens they represent want these things. There is a lot of wind potential in West Virginia and it's ready to be developed. These jobs can be created now.

Always know that many others are working toward the same goals as you are. Ask for help and guidance if you need it, but don't let others define what is best for your community. That is up to you.

ENDNOTES

- ⁱ “Journey Up Coal River” is an interactive mapping project that combines music, audio, photography and the written word to tell the story of the Coal River Valley in West Virginia. The companion CD, [*Still Moving Mountains: The Journey Home*](#), features additional music and interviews with contributions from Kathy Mattea, The Del McCoury Band, and many others.
- ⁱⁱ <http://awea.org/projects/Projects.aspx?s=West+Virginia>
- ⁱⁱⁱ Biomass can also be seen as a renewable fuel as long as the wood source is not extracted faster than it can grow back.
- ^{iv} U.S. Geological Survey Professional Paper 1625–C (USGS) (2001). Chapter A: Executive Summary—Coal Resource Assessment of Selected Coal Beds and Zones in the Northern and Central Appalachian Basin Coal Regions. http://pubs.usgs.gov/pp/p1625c/CHAPTER_A/CHAPTER_A.pdf
- ^v A megawatt (MW) is the same as a million watts, or the equivalent of the energy required to power ten thousand 100-watt light bulbs. A megawatt-hour (MWh) is the amount of electricity generated by a 1 MW generator in a single hour. A 1 MW electricity generator running at full capacity will produce 8,760 MWh over the course of a single year, because there are 8,760 hours in a year. A megawatt is also the same as 1,000 kilowatts (kW).
- ^{vi} This study was conducted by AWS Truewind and can be obtained by contacting the West Virginia Division of Energy.
- ^{vii} Accounting for the fact that the wind doesn’t blow all the time, and that a wind farm will end up producing about 30% of its potential capacity (meaning that a 100 MW wind farm will actually produce the equivalent of a 30 MW coal-fired electricity plant), 3,830 MW of wind development could provide more than 10% of all the electricity generated in West Virginia. Only about one-third of the electricity generated at West Virginia power plants is actually consumed by West Virginia customers.
- ^{viii} See <http://awea.org/projects/Projects.aspx?s=West+Virginia>
- ^{ix} See <http://www.capacity-magazine.com/wmspage.cfm?parm1=383>
- ^x Ibid.
- ^{xi} See www.coalriverwind.org
- ^{xii} A kilowatt (kW) is the same as a thousand watts, or the equivalent of the energy required to power ten 100-watt light bulbs. A kilowatt-hour (kWh) is the amount of electricity generated by a 1 kW generator in a single hour. A 1 kW electricity generator running at full capacity will produce 8,760 kWh over the course of a single year, because there are 8,760 hours in a year. A MW is also the same as 1,000 kW.
- ^{xiii} www.oregoncommunitywind.com/community.html
- ^{xiv} See www.jobs-project.org
- ^{xv} E. Lantz and S. Tegen (2009). Economic Development Impacts of Community Wind Projects. April. www.nrel.gov/docs/fy09osti/45555.pdf
- ^{xvi} This calculation is based upon West Virginia Senate Bill 441 (SB 441). The law says that 12 percent of a generating plant’s maximum capacity is taxable at \$22.78 per kilowatt. So 12 MW of a 100 MW wind farm, or 12,000 kW, is taxed at \$22.78, for a total of approximately \$275,000 per year.
- ^{xvii} Downstream Strategies (2008). The Long-term Economic Benefits of Wind vs. Mountaintop Removal Coal on Coal River Mountain, West Virginia. www.downstreamstrategies.com/Documents/reports_publication/Wind_vs_mountaintop_removal_coal_Coal_River_Mt_n_Dec2008.pdf
- ^{xviii} See <http://www.wvsto.com/dept/Admin/Tax/Pages/CoalSeveranceTax.aspx> for an accounting of the severance taxes returned to local units of government. However, state government retains 93% of all severance taxes.
- ^{xix} Id. Downstream Strategies (2008).
- ^{xx} See http://www.windpoweringamerica.gov/filter_detail.asp?itemid=2074
- ^{xxi} Id. Downstream Strategies (2008).
- ^{xxii} Flowers, Larry. National Renewable Energy Laboratory (NREL) (2009). Wind Energy Update. Slideshow presentation to the West Virginia Wind Working Group, October, 2008. Updated: February, 2009. www.energywv.org/community/WindConferences.html
- ^{xxiii} Ibid.
- ^{xxiv} Id. Downstream Strategies (2008).